

Figure 1: The VLE curve for the ethanol/water mixture

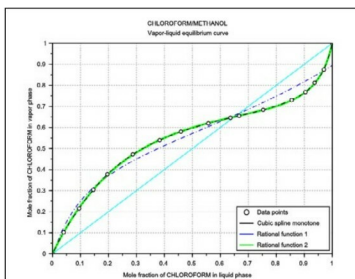


Figure 2: The VLE curve for the chloroform/methanol mixture

```
// Cubic spline monotone
df=spln(x,y,"monotone");
[yyf,yy1f,yy2f]=interp(xx,x,y,df);

plot([0,1],[0,1],"c-"," // Diagonal
      xx,yp1,"b-"," // Rational function 1
      xx,yp2,"g-"," // Rational function 2
      xx,yyf,"k-"," // Cubic spline monotone
      x,y,"ko"); // Data points
```

The Rational function 2 was chosen as the most interesting (see the green curve in Figures 1 and 2) and six coefficients will be used to define each VLE curve. A simple polynomial model was not chosen because of its instability.

The McCabe-Thiele method

The next step is to draw the q-line. This line starts on the diagonal at the x_F value and has a slope which depends on the feed thermal conditions. In Figures 4 and 5, because we have a saturated-liquid feed, the q-line is vertical. For the other q-line types, see the third

and fourth references at the end of this article. Then it's possible to draw another line called the operating line for the enriching section (the section above the feed inlet, the yellow line in Figures 4 and 5). This second line starts on the diagonal at the x_D value and cuts the y-axis at the value $x_D/(R+1)$. The last line is called the operating line for the exhausting section (the section below the feed inlet, the magenta line in Figures 4 and 5). This third line starts on the diagonal at the x_B value and ends at the intersection (x_i, y_i) of the two previous lines.

```
xF=0.35; // Feed mole fraction
xD=0.6; // Distillate mole fraction
xB=0.05; // Bottom product mole fraction
R=1.5; // Reflux ratio R=L/D
q=1; // Feed thermal conditions
xi=(-(q-1)*(1-R/(R+1))*xD-xF)/((q-1)*R/(R+1)-q); //
Intersection x point
yi=(xF+xD*q/R)/(1+q/R); //
Intersection y point
plot([0,1],[0,1],"c-"," // Diagonal
      xx,yp,"k-"," // Rational function 2
      [xF,xi],[xF,yi],"k-"," // q-line
      [xD,xi],[xD,yi],"y-"," // Operating line enriching section
      [xB,xi],[xB,yi],"m-"); // Operating line exhausting section
```

Finally, it's possible to draw the steps between the operating lines and the VLE curve and then count them (Figures 4 and 5). These steps represent the equilibrium stages (theoretical plates) required to carry out the distillation at the given conditions. The required number of theoretical plates is seven for the example in Figure 4, and six for the example in Figure 5. The Scilab code used to plot the operating lines and the equilibrium stages is adapted to Scilab from the MATLAB code written by Housam Binous, as reported in the fifth reference at the end of this article.

```
// VLE curve equation
function f=equilb(x)
    f=y-((coe(1)+coe(2)*x+coe(3)*x.^2)./(1+coe(4)*x+coe(5)*x.^2+coe(6)*x.^3));
endfunction

// Enriching section
i=1;
xp(1)=xD;
yp(1)=xD;
y=xD;
while (xp(i)>xi),
    xp(i+1)=fsolve(0.01,equilib); // Steps x coordinates
    yp(i+1)=R/(R+1)*xp(i+1)+xD/(R+1); // Steps y coordinates
    y=yp(i+1);
    plot([xp(i),xp(i+1)],[yp(i),yp(i+1)],"r-"); // Steps plot
    if (xp(i+1)>xi) then
        plot([xp(i+1),xp(i+1)],[yp(i),yp(i+1)],"r-"); //
```